

Math 526, F08
Quiz 10

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Find an orthogonal matrix Q that diagonalizes $A = \begin{bmatrix} -2 & 6 \\ 6 & 7 \end{bmatrix}$, i.e., $A = Q\Lambda Q^T$.

$$\text{Solution: } \det(A - \lambda I) = \det \begin{bmatrix} -2-\lambda & 6 \\ 6 & 7-\lambda \end{bmatrix} = (-2-\lambda)(7-\lambda) - 36$$

$$= \lambda^2 - 5\lambda - 50$$

$$= (\lambda - 10)(\lambda + 5) = 0$$

$$\Rightarrow \lambda_1 = 10, \lambda_2 = -5$$

$$\text{For } \lambda_1 = 10, (A - 10I)x_1 = 0, \begin{bmatrix} -12 & 6 \\ 6 & -3 \end{bmatrix} x_1 = 0$$

$$\Rightarrow x_1 = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\text{Normalize } x_1, \text{ we get } x_1 = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$$\text{For } \lambda_2 = -5, (A + 5I)x_2 = 0, \begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix} x_2 = 0$$

$$\Rightarrow x_2 = c_2 \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

$$\text{Normalize } x_2, \text{ we get } x_2 = \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

$$\Rightarrow Q = \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} \text{ and}$$

$$A = Q \begin{bmatrix} 10 & \\ & -5 \end{bmatrix} Q^T$$

Problem 2. (10 points) For what numbers c is $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & c & 1 \\ 0 & 1 & c \end{bmatrix}$ positive definite?

Solution: By the determinant test, we need

$$\det[A] = 2 > 0$$

$$\det \begin{bmatrix} 2 & 0 \\ 0 & c \end{bmatrix} = 2c > 0 \Rightarrow \underline{c > 0}$$

$$\det \begin{bmatrix} 2 & 0 & 0 \\ 0 & c & 1 \\ 0 & 1 & c \end{bmatrix} = 2(c^2 - 1) > 0 \Rightarrow c^2 - 1 > 0 \Rightarrow \underline{c > 1 \text{ or } c < -1}$$

$\Rightarrow A$ is positive definite when $c > 1$